

Quizzes & Assessments — Git & GitHub for DevOps

5 Session Quizzes + 1 Final Comprehensive Exam

How to Use These Assessments

- **Quiz time:** 10 minutes per session quiz, 30 minutes for final
 - **Format:** Multiple choice + practical scenarios
 - **Grading:** Immediate self-check (answers at bottom of each section)
 - **Passing:** 70% on quizzes, 80% on final
-
-

SESSION 1 QUIZ: Your First Repository

Time: 10 minutes | **Questions:** 8

Question 1

What does `git init` do?

- A) Downloads code from GitHub
 - B) Creates a new `.git` folder to start tracking a directory
 - C) Commits all files in the folder
 - D) Connects to a remote server
-

Question 2

What is the staging area?

- A) A folder where Git stores old commits
 - B) A temporary holding area for files before they are committed
 - C) The GitHub website
 - D) A backup folder
-

Question 3

Which command puts a file into the staging area?

- A) `git commit`
- B) `git push`

- C) `git add`
 - D) `git init`
-

Question 4

Which command permanently saves your staged files?

- A) `git add`
 - B) `git save`
 - C) `git commit`
 - D) `git stage`
-

Question 5

What happens if you run `git status` in a folder that is NOT a Git repository?

- A) It shows "nothing to commit"
 - B) It shows "fatal: not a git repository"
 - C) It creates a repository automatically
 - D) It lists all files in the folder
-

Question 6

Which of these is a GOOD commit message?

- A) "asdf"
 - B) "fix stuff"
 - C) "Add nginx configuration for production"
 - D) "update"
-

Question 7

What does `git log --oneline` show?

- A) All deleted files
 - B) A compact list of commits
 - C) All branches
 - D) All remote servers
-

Question 8 (Practical Scenario)

You create a file `config.txt` and want to commit it. What is the correct sequence?

Write the commands in order:

1. _____
 2. _____
 3. _____
-

Answers: Session 1

1. **B** — `git init` creates the `.git` folder and starts tracking the directory.
2. **B** — The staging area temporarily holds files before they are committed.
3. **C** — `git add` stages files.
4. **C** — `git commit` saves staged files permanently.
5. **B** — Git shows "fatal: not a git repository" if `.git` doesn't exist.
6. **C** — Descriptive messages explain what changed and why.
7. **B** — `git log --oneline` shows each commit as one line.
8. Solution:
``bash
9. `git add config.txt`
10. `git commit -m "Add configuration file"`

11. git log --oneline

^^

SESSION 2 QUIZ: Branching and Merging

Time: 10 minutes | **Questions:** 8

Question 1

Why do we use branches?

- A) To make Git run faster
 - B) To protect `main` while we experiment safely
 - C) To delete old commits
 - D) To connect to GitHub
-

Question 2

Which command creates a new branch?

- A) `git create branch-name`
 - B) `git branch branch-name`
 - C) `git new branch-name`
 - D) `git switch branch-name`
-

Question 3

Which command switches to an existing branch?

- A) `git branch branch-name`
- B) `git change branch-name`

- C) `git switch branch-name`
 - D) `git move branch-name`
-

Question 4

What is a fast-forward merge?

- A) A merge that deletes the branch being merged
 - B) A merge where `main` hasn't changed since branching — Git just moves the pointer
 - C) A merge that runs faster than normal
 - D) A merge that doesn't require a commit message
-

Question 5

What happens if you delete a branch after merging?

- A) All its commits are lost
 - B) Its commits are safely in `main` ; the branch label is deleted
 - C) The merge is undone
 - D) Git shows an error
-

Question 6

Create and switch to a branch in one command:

- A) `git branch --create branch-name`
 - B) `git switch -c branch-name`
 - C) `git checkout --new branch-name`
 - D) `git branch -s branch-name`
-

Question 7

You switch to `main` and run `ls`. A file from your feature branch is NOT there. Why?

- A) The file was deleted
 - B) Branches are isolated — changes on one branch don't appear on another until merged
 - C) Git is broken
 - D) The file is hidden
-

Question 8 (Practical Scenario)

You need to add a new `contact.txt` page via a branch. Write the complete sequence:

1. _____
 2. _____
 3. _____
 4. _____
 5. _____
 6. _____
 7. _____
-

Answers: Session 2

1. **B** — Branches let you work safely without affecting `main`.
2. **B** — `git branch branch-name` creates a branch.
3. **C** — `git switch branch-name` changes branches.
4. **B** — Fast-forward happens when `main` hasn't diverged.
5. **B** — Commits stay in history; only the branch label is deleted.
6. **B** — `git switch -c branch-name` creates and switches.
7. **B** — Branches are isolated by design.

8. Solution:

```
```bash
```

9. git branch feature-contact

10. git switch feature-contact

11. echo "Email: hello@example.com" > contact.txt

12. git add contact.txt

13. git commit -m "Add contact page"

14. git switch main

15. git merge feature-contact

16. git branch -d feature-contact

```
```
```

SESSION 3 QUIZ: GitHub, Push, Pull, and Pull Requests

Time: 10 minutes | **Questions:** 8

Question 1

What is a remote?

- A) A git command that undoes changes
 - B) A copy of your repository on another server (like GitHub)
 - C) A branch that is far away from `main`
 - D) A deleted commit
-

Question 2

Which command uploads your commits to GitHub?

- A) `git upload`
 - B) `git send`
 - C) `git push`
 - D) `git publish`
-

Question 3

Which command downloads the latest changes from GitHub?

- A) `git download`
- B) `git fetch`

- C) `git pull`
 - D) `git receive`
-

Question 4

What does `git remote add origin ...` do?

- A) Deletes the remote
 - B) Creates a nickname `origin` for your GitHub repo URL
 - C) Pushes code to GitHub
 - D) Creates a new branch called `origin`
-

Question 5

What is a Pull Request?

- A) A request to delete a branch
 - B) A formal request to merge your branch into main, with code review
 - C) A command to download code
 - D) An email to your manager
-

Question 6

Why shouldn't you push directly to `main` ?

- A) It's faster to use branches
 - B) Direct pushes bypass code review and can break production
 - C) GitHub doesn't allow it
 - D) Branches cost money
-

Question 7

What does `git clone` do?

- A) Deletes a repository
 - B) Copies a remote repository to your local machine
 - C) Creates a new empty repository
 - D) Merges two branches
-

Question 8 (Practical Scenario)

You just merged a PR on GitHub. Your local `main` doesn't have the changes yet. Write the commands to sync:

1. _____
 2. _____
 3. _____
-

Answers: Session 3

1. **B** — A remote is a copy on a server.
2. **C** — `git push` uploads commits.
3. **C** — `git pull` downloads and merges; `git fetch` only downloads.
4. **B** — `origin` is a nickname for the remote URL.
5. **B** — PRs request merging with review.
6. **B** — Direct pushes skip quality checks.
7. **B** — `git clone` copies a remote repo locally.
8. Solution:
```bash
9. `git switch main`
10. `git pull`

11. `git branch -d feature-branch-name`  
``

---

---

# SESSION 4 QUIZ: Undoing and Conflicts

---

**Time:** 10 minutes | **Questions:** 8

---

## Question 1

---

You need to undo a bad commit that was already pushed. What do you use?

- A) `git reset --hard`
  - B) `git revert`
  - C) `git delete`
  - D) `git erase`
- 

## Question 2

---

What does `git reset --soft HEAD~1` do?

- A) Deletes all files
  - B) Removes the last commit but keeps the changes staged
  - C) Deletes the repository
  - D) Pushes to GitHub
- 

## Question 3

---

You see this in a file:

```
<<<<<<< HEAD
my version
=====
their version
>>>>>>> their-branch
```

What is this?

- A) A bug in Git
  - B) A merge conflict that needs manual resolution
  - C) A virus
  - D) An encrypted message
- 

## Question 4

---

After fixing a merge conflict, what must you do?

- A) Delete the file
  - B) Run `git restart`
  - C) `git add .` then `git commit`
  - D) Run `git fix`
- 

## Question 5

---

What is `git stash` used for?

- A) Deleting branches
  - B) Saving uncommitted work temporarily so you can switch tasks
  - C) Pushing to GitHub
  - D) Creating a new repository
-



- 5. **B** — Stash saves work-in-progress temporarily.
  - 6. **D** — `pop` applies and removes; `apply` applies but keeps it.
  - 7. **B** — Reflog records everything — your safety net.
  - 8. Solution: `git reset --soft HEAD~1`
- 
-

# SESSION 5 QUIZ: CI/CD with GitHub Actions

---

**Time:** 10 minutes | **Questions:** 8

---

## Question 1

---

What does CI stand for?

- A) Code Integration
  - B) Continuous Integration
  - C) Computer Intelligence
  - D) Central Integration
- 

## Question 2

---

GitHub Actions workflows are stored in which folder?

- A) `.github/actions/`
  - B) `.github/workflows/`
  - C) `.actions/`
  - D) `workflows/`
- 

## Question 3

---

What file format are GitHub Actions workflows written in?

- A) JSON
- B) XML

- C) YAML
  - D) HTML
- 

## Question 4

---

What does `runs-on: ubuntu-latest` mean?

- A) Your laptop must run Ubuntu
  - B) GitHub will use a virtual machine running Ubuntu
  - C) The workflow only works on Ubuntu
  - D) The branch name must be `ubuntu-latest`
- 

## Question 5

---

What does `actions/checkout@v4` do?

- A) Deletes the repository
  - B) Downloads your code onto the runner
  - C) Creates a new branch
  - D) Logs into GitHub
- 

## Question 6

---

What is branch protection?

- A) A firewall around your code
  - B) Rules that prevent direct pushes and enforce PRs + CI
  - C) A password on the repository
  - D) A backup service
-

## Question 7

---

Where do you store secrets (like API keys) in GitHub Actions?

- A) In the workflow file
  - B) In a text file in the repo
  - C) In Repo Settings → Secrets and variables
  - D) In the README
- 

## Question 8 (Practical Scenario)

---

Write a workflow that triggers on every push to `main` and prints "Deploying to production!".

```
name: _____
on:
 _____:
 branches: [_____]
jobs:
 deploy:
 runs-on: _____
 steps:
 - run: echo "Deploying to production!"
```

---

## Answers: Session 5

---

1. **B** — Continuous Integration
2. **B** — `.github/workflows/`
3. **C** — YAML
4. **B** — GitHub provides a VM (runner) running Ubuntu
5. **B** — It checks out your code onto the runner
6. **B** — Rules for `main` : PRs required, CI must pass
7. **C** — Secrets belong in Settings → Secrets, NEVER in code

8. Solution:

```
yaml name: Deploy on: push: branches: [main] jobs: deploy: runs-on:
ubuntu-latest steps: - run: echo "Deploying to production!"
```

---

---

# FINAL COMPREHENSIVE EXAM

---

**Time:** 30 minutes | **Questions:** 15 | **Passing:** 80%

---

## Section A: Multiple Choice (10 questions)

---

### Question 1

Which command initializes a new Git repository?

- A) `git start`
- B) `git init`
- C) `git new`
- D) `git create`

### Question 2

What is the correct sequence?

- A) `add` → `commit` → `push`
- B) `commit` → `add` → `push`
- C) `push` → `add` → `commit`
- D) `init` → `push` → `commit`

### Question 3

You want to undo a commit that is already on GitHub. You should:

- A) `git reset --hard`
- B) `git revert`
- C) Delete the repository
- D) Email GitHub support

## Question 4

Two branches edited the same line. What does Git do?

- A) Automatically picks the first branch
- B) Creates a merge conflict and waits for a human
- C) Deletes both changes
- D) Emails the team

## Question 5

What does `git stash pop` do?

- A) Deletes all stashes
- B) Applies the latest stash and removes it
- C) Shows stash history
- D) Pushes code to GitHub

## Question 6

A Pull Request is:

- A) A request to delete a branch
- B) A formal request to merge code with review
- C) A command to download code
- D) An email to the manager

## Question 7

Where are GitHub Actions workflows stored?

- A) `.github/actions/`
- B) `.github/workflows/`
- C) `actions.yml`
- D) `ci.yml`

### Question 8

What does `git push -u origin main` do?

- A) Deletes the main branch
- B) Pushes main and sets upstream tracking
- C) Creates a new repository
- D) Pulls code from GitHub

### Question 9

Which is the safe way to undo a local commit while keeping changes?

- A) `git reset --hard HEAD~1`
- B) `git reset --soft HEAD~1`
- C) `git rm -rf .`
- D) `git undo`

### Question 10

What does branch protection prevent?

- A) Creating branches
- B) Direct pushes to main without PR or CI
- C) Opening pull requests
- D) Deleting branches

---

## Section B: Practical Scenarios (5 questions)

---

### Question 11

Write the complete sequence to set up a new repo, add a file, and push it to GitHub.

---

---

---

---

---

---

---

## Question 12

You committed a file to `main` but meant to commit to a feature branch. Write the fix.

---

---

---

## Question 13

Write a GitHub Actions workflow that runs on every pull request to `main` and prints "Running tests...".

---

---

---

---

---

---

---

---

---

## Question 14

You see a merge conflict in `config.txt`. Write the exact steps to resolve it.

1. 

---
2. 

---
3. 

---
4. 

---

### Question 15 (Bonus)

You accidentally committed a file called `secrets.env` containing a password. It has NOT been pushed. Write the complete fix.

1. \_\_\_\_\_
2. \_\_\_\_\_
3. \_\_\_\_\_
4. \_\_\_\_\_
5. \_\_\_\_\_

## Final Exam Answers

### Section A

| Q  | Answer |
|----|--------|
| 1  | B      |
| 2  | A      |
| 3  | B      |
| 4  | B      |
| 5  | B      |
| 6  | B      |
| 7  | B      |
| 8  | B      |
| 9  | B      |
| 10 | B      |

### Section B

Question 11:

```
git init
git add .
git commit -m "Initial commit"
git remote add origin git@github.com:USER/REPO.git
git branch -M main
git push -u origin main
```

### Question 12:

```
git switch -c feature-branch # Brings the commit
git switch main
git reset --soft HEAD~1
```

### Question 13:

```
name: Test
on:
 pull_request:
 branches: [main]
jobs:
 test:
 runs-on: ubuntu-latest
 steps:
 - run: echo "Running tests..."
```

### Question 14:

1. Open config.txt, remove <<<<<< ===== >>>>>> markers
2. Edit to keep desired content
3. git add config.txt
4. git commit -m "Resolve config conflict"

### Question 15 (Bonus):

1. git reset --soft HEAD~1 # Undo last commit, keep changes staged
  2. git restore --staged secrets.env # Unstage the file
  3. echo "secrets.env" >> .gitignore # Add to .gitignore
  4. git add .gitignore
  5. git commit -m "Add .gitignore and remove secret"
- # Then change the password immediately!

## Scoring Guide

---

| Score | Grade | Meaning                 |
|-------|-------|-------------------------|
| 15/15 | A+    | DevOps ready            |
| 13-14 | A     | Strong understanding    |
| 11-12 | B     | Good, review weak areas |
| 9-10  | C     | Needs more practice     |
| < 9   | D/F   | Retake Session quizzes  |

---

*Good luck!*